

**This assignment will not be graded and is only for practice.**

**Multiple Choice Questions (MCQ):**

1) Which of the following statements are correct?

**1 point**

☐ There exists a linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  such that  $T(2, 3) = (1, 2)$ ,  $T(1, -1) = (1, -1)$ , and  $T(4, 1) = (1, 0)$ .

☐ If there is a linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  such that  $T(2, 3) = (1, 2)$ , and  $T(1, -1) = (1, -1)$ , then  $T(x, y) = \frac{1}{5}(4x - y, -x + 4y)$ .

Let  $\beta = \{v_1, v_2, v_3\}$  and  $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$  be bases of a vector space  $V$ . Consider a linear transformation  $T : V \rightarrow V$  such that  $T(v_1) = 2v_3 + v_1$ ,  $T(v_2) = 2v_1$ , and  $T(v_3) = 2v_2$ . The matrix representation of  $T$ , with respect to  $\beta$  and  $\gamma$  for the domain and codomain respectively, is

☐ 
$$\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}$$

Let  $\{v_1, v_2, v_3\}$  be a basis of a vector space  $V$  and  $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$  be a basis of vector space  $W$ . Consider a linear transformation  $T : V \rightarrow W$  such that  $T(v_1) = 2v_3 + v_1$ ,  $T(v_2) = 2v_1$ , and  $T(v_3) = 2v_2$ . Then, the rank of  $T$  is 3.

☐

No, the answer is incorrect.

Score: 0

Accepted Answers:

If there is a linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  such that  $T(2, 3) = (1, 2)$ , and  $T(1, -1) = (1, -1)$ , then  $T(x, y) = \frac{1}{5}(4x - y, -x + 4y)$ .

Let  $\beta = \{v_1, v_2, v_3\}$  and  $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$  be bases of a vector space  $V$ . Consider a linear transformation  $T : V \rightarrow V$  such that  $T(v_1) = 2v_3 + v_1$ ,  $T(v_2) = 2v_1$ , and  $T(v_3) = 2v_2$ . The matrix representation of  $T$ , with respect to  $\beta$  and  $\gamma$  for the

domain and codomain respectively, is 
$$\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}$$

Let  $\{v_1, v_2, v_3\}$  be a basis of a vector space  $V$  and  $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$  be a basis of vector space  $W$ . Consider a linear transformation  $T : V \rightarrow W$  such that  $T(v_1) = 2v_3 + v_1$ ,  $T(v_2) = 2v_1$ , and  $T(v_3) = 2v_2$ . Then, the rank of  $T$  is 3.

2) Consider the following set  $S = \{(x, y) \mid x + y = 1, x, y \in \mathbb{R}\}$ . In column A the matrix representation of some linear transformations are given with respect to the standard ordered bases. Match the entries in column A with the set  $T(S)$  given in column B and the

**1 point**

geometric representations of  $S$  and  $T(S)$  in column C.

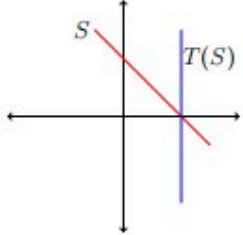
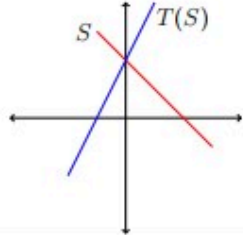
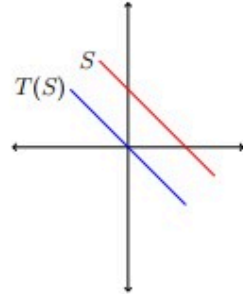
|      | Matrix form of linear transformation<br>(Column A)                                              |    | Image of the given set<br>(Column B)                      |    | Geometric representations<br>(Column C)                                               |
|------|-------------------------------------------------------------------------------------------------|----|-----------------------------------------------------------|----|---------------------------------------------------------------------------------------|
| i)   | $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$<br>$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ | a) | $T(S) = \{(x, y) \mid x = 1, y \in \mathbb{R}\}$          | 1) |    |
| ii)  | $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$<br>$\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$ | b) | $T(S) = \{(x, y) \mid x + y = 0, x, y \in \mathbb{R}\}$   | 2) |   |
| iii) | $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$<br>$\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$  | c) | $T(S) = \{(x, y) \mid 2x - y = -1, x, y \in \mathbb{R}\}$ | 3) |  |

Table: M2W6P1

- ☐ i  $\rightarrow$  b  $\rightarrow$  2.
- ☐ ii  $\rightarrow$  c  $\rightarrow$  1.
- ☐ ii  $\rightarrow$  c  $\rightarrow$  2.
- ☐ iii  $\rightarrow$  a  $\rightarrow$  3.

☐  $i \rightarrow b \rightarrow 3.$

☐  $iii \rightarrow a \rightarrow 1.$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$ii \rightarrow c \rightarrow 2.$

$i \rightarrow b \rightarrow 3.$

$iii \rightarrow a \rightarrow 1.$

**Multiple Select Questions (MSQ):**

3) Consider the following two groups. In Group 1, there are 3 functions from one vector **1 point** space to another, and in Group 2, there are three properties of functions. All the vector spaces have usual addition and scalar multiplication.

**Group-1**

$$\mathbf{P}: T : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ such that } T(x, y) = \begin{cases} (\frac{x^2}{y}, y) & \text{if } y \neq 0 \\ (x, 0) & \text{otherwise} \end{cases}$$

$$\mathbf{Q}: T : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ such that } T(x, y) = (x, xy)$$

$$\mathbf{R}: T : \mathbb{R}^3 \rightarrow \mathbb{R} \text{ such that } T(x, y, z) = \frac{x+y+z}{5}$$

**Group-2**

$$1: T(v_1 + v_2) = T(v_1) + T(v_2) \text{ for all } v_1, v_2 \in V$$

$$2: T(cv) = cT(v) \text{ for all } v \in V \text{ and } c \in \mathbb{R}.$$

3:  $T$  is a linear transformation

Choose the set of correct options.

- ☐ P satisfies 1 but not 2.
- ☐ P satisfies 2 but not 1
- ☐ Q satisfies 1
- ☐ Q satisfies 3
- ☐ P satisfies 3
- ☐ R satisfies 2
- ☐ R satisfies 3
- ☐ R satisfies 1 but not 2

No, the answer is incorrect.

Score: 0

Accepted Answers:

P satisfies 2 but not 1

R satisfies 2

R satisfies 3

**Numerical Answer Type (NAT):**

4) Suppose the matrix representation of a linear transformation  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  with respect to the ordered bases  $\beta = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$  for the domain and  $\gamma = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$  for the range, is  $I_{3 \times 3}$ , i.e., the identity matrix of order 3. Let  $A$  be the matrix representation of the linear transformation  $T$  with respect to standard ordered basis of  $\mathbb{R}^3$  for both domain and range. Then the determinant of  $A$  is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

**1 point**

5) Let  $T$  and  $S$  be two linear transformations from  $\mathbb{R}^3$  to  $\mathbb{R}^3$ , which are defined as follows:

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3 \quad S : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (x - y, y - z, z - x) \quad S(x, y, z) = (x + y, y + z, z + x)$$

Let  $S + T$  be defined to be the linear transformation as follows:

$$(S + T) : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$(S + T)(x, y, z) = S(x, y, z) + T(x, y, z)$$

Let  $C$  be the matrix representation of  $S + T$  with respect to the standard ordered bases of  $\mathbb{R}^3$ . If  $C = nI$ , where  $I$  denotes the identity matrix of order 3, then what will be the value of  $n$ ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

6) Consider a linear transformation  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  defined as  $T(x, y, z) = (x + 5y, z - 3x, y + 6z)$ . Which of the following options are true about  $T$ ? **1 point**

- ☐ Nullity of  $T$  is 0.
- ☐ A basis for the range of  $T$  is  $\{(1, 5, 0), (0, 1, 6)\}$ .
- ☐ A basis for the range of  $T$  is  $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ .
- ☐ Rank of  $T$  is 2.

The matrix representation of  $T$  with respect to the standard ordered bases of

☐  $\mathbb{R}^3$  is  $\begin{bmatrix} 1 & -3 & 0 \\ 5 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$ .

The matrix representation of  $T$  with respect to the standard ordered bases of

☐  $\mathbb{R}^3$  is  $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$ .

- ☐ Nullity of  $T$  is 1.
- ☐ Rank of  $T$  is 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Nullity of  $T$  is 0.

A basis for the range of  $T$  is  $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ .

The matrix representation of  $T$  with respect to the standard ordered bases of  $\mathbb{R}^3$  is

$\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$ .

Rank of  $T$  is 3.

7) Let  $V$  be a vector space with ordered basis  $\alpha = \{v_1, v_2, \dots, v_{10}\}$ . Consider a linear transformation  $T : V \rightarrow V$  such that  $T(v_1) = v_1$  and  $T(v_i) = v_i - v_{i-1}$ , where  $i > 1$ . Let  $A$  be the matrix representation of the linear transformation  $T$ , with respect to  $\alpha$  for both the domain and codomain. what is the trace of  $A$ ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 10

1 point

### Comprehension Type Question:

For each video on YouTube, consider the vector (Number of comments in thousands, Number of shares in thousands, Number of views in thousands) and consider the subset  $S$  of  $\mathbb{R}^3$  consisting of these vectors. It is known that the ranking of a YouTube video can be thought of as the restriction of a linear transformation  $T : \mathbb{R}^3 \rightarrow \mathbb{R}$  to  $S$  where  $\mathbb{R}^3$  comes with usual addition and scalar multiplication. (assume ranking 0 (zero) is possible and in practice negative rankings can be ignored).

The following data in Table M2W6P2 shows the correlation between comments, shares, and views - with YouTube rankings of a video.

| Number of comments<br>(in thousands) | Number of shares<br>(in thousands) | Number of views<br>(in thousands) | Ranking of a video |
|--------------------------------------|------------------------------------|-----------------------------------|--------------------|
| 3                                    | 10                                 | 8                                 | 3                  |
| 4                                    | 15                                 | 10                                | 2                  |
| 5                                    | 20                                 | 15                                | 1                  |

Table: M2W6P2

Use the above information to answer questions 8, 9 and 10.

8) Which of the following sets of vectors in  $S$  have the property that the YouTube video **1 point** corresponding to every linear combination of that set which is in  $S$  has ranking 0?

- ☐  $\{(1, 1, 0), (0, 0, 1)\}$
- ☐  $\{(6, 25, 0), (0, 0, 25)\}$
- ☐  $\{(25, 6, 0), (0, 0, 25)\}$
- ☐  $\{(12, 50, 0), (0, 0, 1)\}$
- ☐  $\{(24, 100, 0), (0, 0, 25)\}$
- ☐  $\{(6, 0, 0), (0, 25, 0), (0, 0, 1)\}$
- ☐  $\{(25, 0, 0), (0, 6, 0), (0, 0, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(6, 25, 0), (0, 0, 25)\}$   
 $\{(12, 50, 0), (0, 0, 1)\}$   
 $\{(24, 100, 0), (0, 0, 25)\}$

9) Which of the following sets of vectors in  $S$  have the property that the YouTube video **1 point** corresponding to every linear combination of that set which is in  $S$  has ranking a multiple of 15?

- ☐  $\{(3, 0, 0), (0, 0, 1)\}$
- ☐  $\{(3, 0, 1)\}$
- ☐  $\{(3, 0, 1), (0, 0, 1)\}$
- ☐  $\{(0, 0, 1)\}$
- ☐  $\{(9, 25, 0), (0, 0, 1)\}$
- ☐  $\{(25, 9, 0), (0, 0, 1)\}$
- ☐

$\{(9, 0, 0), (0, 25, 0), (0, 0, 1)\}$

☐  $\{(25, 0, 0), (0, 9, 0), (0, 0, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(3, 0, 0), (0, 0, 1)\}$

$\{(3, 0, 1)\}$

$\{(3, 0, 1), (0, 0, 1)\}$

$\{(0, 0, 1)\}$

$\{(9, 25, 0), (0, 0, 1)\}$

$\{(9, 0, 0), (0, 25, 0), (0, 0, 1)\}$

10) Suppose that for a YouTube video there are 5 (in thousands) shares. What must be the number of comments (in thousands) so that the ranking of the video will be 4? [Note: Suppose your answer is 6000, then you have to enter 6 as the answer.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

Your score is: 0/10